

Retrieval-Augmented Generation: A Study

Abstract

This paper explores retrieval-augmented generation methods for reducing hallucination in large language models. We survey the current landscape and propose a new evaluation framework. Our contributions include an open benchmark, a reproducible evaluation harness, and a detailed error analysis.

1. Introduction

Large language models have shown remarkable capabilities but suffer from hallucination. Grounding outputs in retrieved evidence has emerged as a leading mitigation strategy. Prior work has demonstrated substantial gains on open-domain question answering, fact verification, and long-form summarization tasks when generation is conditioned on retrieved evidence.

2. Methods

We describe a two-stage retrieve-then-read pipeline operating over a dense passage index, with top-k passages fused into the generator context via an explicit provenance marker. Our evaluation harness logs grounding scores, citation coverage, and redundancy metrics per query.
